

Abhay Singh Raihal

Indian Institute of Technology, Mandi
B.Tech - Data Science & Engineering

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abhay-singh-raihal in

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Mandi	8.13	2020-2024
Senior Secondary	CBSE Board	95.20%	2020
Secondary	CBSE Board	89.40%	2018

Relevant Coursework : Data Structure & Algorithms, OOPs, Operating Systems, Information & Database Systems, Mathematical Foundations of Computer Science, Mobile Virtual Reality and AI, Machine Learning, Data Science, Deep Learning.

WORK EXPERIENCE

- **SDE1, Limeroad**  Nov 2024 - Present
Backend, Java Spring MySQL Mongo Gurgaon, Haryana
 - **Optimized edge case handling** for the mcout lead generation system, improving error detection and resolution for users abandoning cart or checkout flows, which enhanced user experience, increased lead accuracy, and ultimately boosted revenue.
 - **Integrated BlueDart as a logistics partner** by developing robust APIs for waybill creation, cancellation, and status tracking using both push and pull mechanisms, ensuring real-time updates and seamless order tracking.
- **SDE Intern, Razorpay**  June - August 2023
Backend: X-Payouts Team Bangalore, Karnataka
 - **Developed and Optimized Admin Actions & Payout Code-flow Analysis:** Designed and deployed production-ready admin actions for specific use cases, resulting in streamlined API integration. Mapped and documented the payout creation route's code-flow, enhancing the understanding of transaction states and improving efficiency.
 - **Implemented and Tested Critical Functions, Enhanced Security:** Optimized Redis operations(GET, SET, and queue management), improving data retrieval speed by **25%**. Introduced comprehensive authorization checks, strengthening security and reducing unauthorized access incidents.

PROJECTS

- **AI Trip Planner**   July-2024
Fullstack Web Application
 - Built a full stack web app using **React**, **Gemini AI**, and **Firebase** to generate personalized trip itineraries, with secure **Google Authentication**, real-time database management, and **Google APIs** for location-based suggestions and media display.
 - Planned a fully responsive interface using **TailwindCSS** and **Shadcn/UI**, deployed on **Vercel** for scalability, and integrated **Pixabay API** to fetch and display images for an enhanced user experience.
- **LightsOut**   January-2023
LAMP stack Web Application
 - Engineered a dynamic movie rating platform using **PHP** and **MySQL**, with secure user authentication, real-time movie reviews, and genre-based filtering to boost user engagement.
 - Designed an intuitive admin panel for efficient management of movie entries, ratings & reviews, incorporating **JavaScript** for real-time updates and **CSS** for a responsive user interface.
- **Self-Driving Car**   May-2024
Autonomous Vehicle Simulation
 - Developed a self-driving car model using **Neural Networks** and **Genetic Algorithm**, coded entirely from scratch in **JavaScript** for autonomous navigation in complex traffic environments, effectively simulating real-world self-driving scenarios.
 - Integrated **real-time local storage** for persistent model retention, supporting both **manual and automated driving modes** to allow users to test, train, and refine models seamlessly across sessions.

SKILLS

- **Programming Languages/ Frameworks:** C/C++, Python, Java, Javascript, React, SQL, GeminiAI
- **Other Tools/ Relevant Libraries:** MongoDB, Firebase, Unity, Latex, Git, Linux, ScikitLearn

POSITION OF RESPONSIBILITY

- Conducted comprehensive surveys for the ISTP project, gathering and analyzing data on local community problems.
- Led & participated in rides organized by Mountain Biking Club (MTB) at IIT Mandi, demonstrating leadership & teamwork.
- Led Design Practicum(DP) project, developing assistive wearable technology to enhance mobility for visually impaired individuals.

ACHIEVEMENTS

- Co-authored the research paper "Attend, Distill, Detect" which was accepted at the prestigious ICPR 2024 conference. 
- Guided a team to attain **1st place** overall in the AI Innovation 3.0 Hackathon at IIT Mandi.
- Led a team to secure **2nd place** overall in Frosthack 2022, a national-level hackathon organized by IIT Mandi.
- Achieved the rank of **Expert**  on Codeforces. Solved **2000+** problems on online coding platforms.